

Question			Answer/Indicative content	Marks	Guidance
1			C	1	
			Total	1	
2	a		* Level 3 (5–6 marks) All points E1, 2, 3 and 4 for equipment All points M1, 2, 3 and 4 for measurements For calculations expect C1, C2, C3 and C4 Expect at least two points from reliability <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Expect E1 and E2; E3 or E4 for equipment Expect M2 and two from M1, M3, M4 for measurements For calculations expect at least C3 and C4 Expect at least one point from reliability <i>There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Expect at least E1 and E2 for equipment Expect at least two from measurements Expect C5 for the calculation No real ideas for obtaining reliable results <i>The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear.</i> 0 marks No response or no response worthy of credit.	B1 × 6	The complete plan consists of four parts: Equipment used safety (E) 1 Wire fixed at one end with load added to wire. 2 Suitable scale with suitable marker on wire. 3 Micrometer screw-gauge or digital / vernier callipers for measuring diameter of wire. 4 Reference to safety concerning wire snapping. Measurements (M) 1 Original length from fixed end to marker on wire. 2 Diameter of wire. 3 Measure load. 4 New length of wire when load increased. Calculation of Young modulus. (C) 1 Find extension (for each load) or strain (for each load). 2 Determine cross-sectional area or stress. 3 Plot graph of load-extension or graph of stress-strain. 4 Young modulus = gradient × original length / area or Young modulus = gradient. 5 Calculate Young modulus from single set of measurements of load, extension, area and length. Reliability of results (R) 1 Measure diameter in 3 or more places and take average. 2 Put on initial load to tension wire and take up 'slack' before measuring original length. 3 Take measurements of extension

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					while unloading to check elastic limit has not been exceeded. 4 Use long wire (to give measurable extension). Scale or ruler parallel to wire.
	b	i	Elastic: material returns to original dimensions when load is removed.	B1	
		i	Plastic: material has permanent change of shape when load is removed.		
		ii	The material is elastic because the removal of force returns the rubber to its original length.	B1	
		ii	The area under force-extension graph is work done.	B1	
		ii	Repeated stretching and releasing the rubber warms up the rubber because not all the strain energy is returned back. The area enclosed represents the amount of thermal energy. During landing, some of the aeroplane's kinetic energy is transferred to thermal energy and therefore the aeroplane does not "bounce" during landing; hence this minimises the risk to passengers. Total	B1	Mentioning 'hysteresis' is not enough to gain this mark.
			D	10	
3			Total	1	
			A	1	
4			Total	1	
			C	1	
5			Total	1	
				1	

Question			Answer/Indicative content	Marks	Guidance
6	a	i	$a = (-) 4\pi^2 f^2 x = 4 \times 9.87 \times 4900 \times 0.004$	C1	allow $774 \text{ (m s}^{-2}\text{)}$
		i	$a = 770 \text{ (m s}^{-2}\text{)}$	A1	
		ii	1 sketch showing one wavelength and 140 (Hz)	B1	both sketch and value required for 1 mark
		ii	2 driving force is around nodal point / AW;	B1	max 3 of the 4 marking points
		ii	points either side of nodal point try to move in opposite directions when force in one direction / AW;	B1	
		ii	move magnet to antinodal point; $\frac{1}{4}$ of distance between clamps	B1	not increase current
	b	i	$f \propto \sqrt{T}$ so $f = 70/\sqrt{2} = 49$ or 50 Hz	B1	
		i	1μ increases / goes up by 0.4%		
		ii		B1	allow +0.4% NOT 0.4%
		ii	2 0.2%,	B1	or half of answer to (ii)1
		ii	f is lower because μ is bigger and μ is on the bottom of the formula	B1	or greater inertia present with same restoring force / other physical argument
			Total	10	
7			B	1	
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
8	a		The extension of each spring is halved because the force in each spring is halved. (Hence the force constant is $2k$.)	B1	Allow $F = kx$, x is halved for the same F , hence k doubles.
	b	i	Missing data point and error bar plotted correctly. Force measured by pulling back plate with	B1	Allow $\frac{1}{2}$ square tolerance.
		ii	a newton-meter. Extension measured with a ruler (placed close to the transparent plastic tube).	B1	
		ii	Best fit line drawn correctly and gradient determined correctly.	B1	
		iii	Worst fit line drawn correctly and its gradient determined correctly.	B1	Ignore POT for this mark; gradient = 50 ± 4 (N m $^{-1}$)
		iii		B1	Note: The line must have a greater/smaller gradient than the best fit line and must pass through all the error bars. Ignore POT for this mark. Possible ECF.
		iii	$2k = 50$ (N m $^{-1}$), therefore $k = 25$ (N m $^{-1}$)	B1	Possible ECF within calculation.
		iii	Absolute uncertainty determined correctly.	B1	
		iv	$F \propto x$ / straight line passing through the origin. energy stored = $\frac{1}{2} \times 50 \times 0.122$	B1	
		v	$\frac{1}{2} \times 50 \times 0.122 = \frac{1}{2} \times 0.39 \times v^2$	C1	Possible ECF from (iii)
		v		C1	
		v	$v = 1.4$ (m s $^{-1}$)	A1	Allow 1 mark for $v = 0.96$ m s $^{-1}$; used k for single spring
	c		force constant of spring arrangement) = $\frac{2k}{3}$ $\frac{2k}{3}x = ma$	M1	
			$a = \frac{2}{3 \times 0.39}kx$	M1	
			$a = 1.7\ kx$	A0	
			Total	14	

Question			Answer/Indicative content	Marks	Guidance
9		i	(Sum of clockwise moments = sum of anticlockwise moments) $95 \times 9.81 \times 1.80 / 120 \times 9.81 \times 1.00 / 1.60 \times T \sin 30^\circ$ $(95 \times 9.81 \times 1.80) + (120 \times 9.81 \times 1.00) = 1.60 \times T \sin 30^\circ$ $T = 3.6 \times 10^3 \text{ (N)}$	C1	
		i		C1	
		i		A1	Note answer to 3 s.f. is $3.57 \times 10^3 \text{ (N)}$
		ii	$\sigma = \frac{3.6 \times 10^3}{\pi \times 0.015^2}$	C1	Possible ECF from part (i)
		ii	$\sigma = 5.1 \times 10^3 \text{ (kPa)}$	A1	Allow 1 mark for 5.1×10^6 ; POT error Note using $3.57 \times 10^3 \text{ N}$ gives $5.05 \times 10^3 \text{ (kPa)}$
		iii	The clockwise moment increases and therefore T increases.	B1	
			Total	6	
10		i	$a = 4\pi^2 f^2 \times$	C1	condition for SHM
		i	so $k = (m4\pi^2 f^2) = 1.7 \times 10^{-27} \times 4 \times 9.87 \times 43.7 \times 10^26$ $k = 292 \text{ (N m}^{-1}\text{)}$	B1	substitution
		i		A1	ecf if incorrect mass used
		ii	(N2 gives) $F_H = m_H a_H$ and $F_I = m_I a_I$	B1	allow total momentum = 0 at all times
		ii	(N3 gives) $F_H = F_I$ <i>can be implicit</i>	B1	SHM gives $v = 2\pi f x_{\text{max}}$
		ii	SHM gives $a \propto (-)x$	B1	so $m_H x_H = m_I x_I$
		ii	hence $x_H/x_I = a_H/a_I = m_I/m_H = 127$	B1	accept $127 = x_H/x_I \approx 10/0.08 = 125$
			Total	7	
			B		
11				1	Examiner's Comments This question was slightly more challenging.
			Total	1	
12			B	1	Examiner's Comments This question proved particularly straightforward and accessible to nearly all candidates.
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
13			A	1	Examiner's Comments This question proved particularly straightforward and accessible to nearly all candidates.
			Total	1	

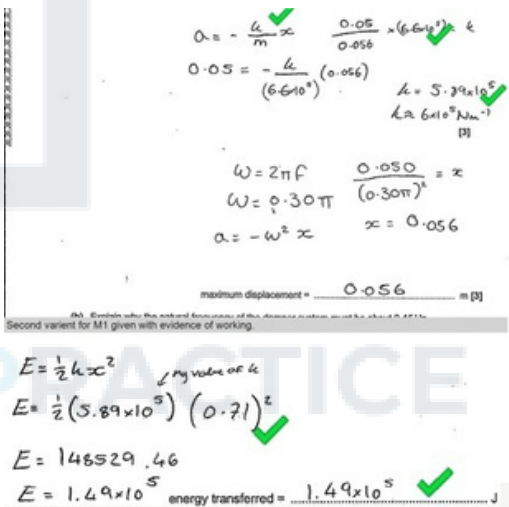


EXAM PAPERS PRACTICE

Question			Answer/Indicative content	Marks	Guidance
14	a		arrow down through centre of ball labeled weight or W or mg or 1.2 N	B1	zero if any other arrows or forces present Examiner's Comments There were some carelessly drawn arrows on the diagram but otherwise this was done well. There were some arrows labelled <i>centripetal force</i> .
	b	i	(horizontally) mv^2/r (or $mr\omega^2$) = $T \sin \theta$ and (vertically) W or mg = $T \cos \theta$ $(\tan \theta = v^2/rg$ or $rw^2/g)$ $\tan \theta = 0.045 \times 4 \times 9.87 \times 2.2 / 9.81$ or $0.48 / 1.2 (= 0.40)$ $\theta = 22^\circ$	M1 A1 A0	accept figures in place of algebra, $r = 0.045$ m $v = 0.42$ m s ⁻¹ $\omega = 3\pi$ rad s ⁻¹ ; $rw^2 = 4.0$ m s ⁻² ; $W = 1.2$ N and $m = 0.12$ kg and $mr\omega^2 = 0.48$ N accept labelled triangle of forces diagram N.B. this is a <i>show that Q</i> ; sufficient calculation must be present to indicate that the candidate has not worked back from the answer
		ii	$k = (mg / x_0 = 1.2 / 0.050) = 24(\text{Nm}^{-1})$ ($T = mg / \cos \theta = kx$ giving) $x = 1.2 / 24 \cos 22$ $x = 0.054$ (m)	C1 C1 A1	or solution by ratios Examiner's Comments About half of the candidates completed the angle calculation successfully with a slightly smaller number finding the correct extension of the string.
	c		$(y = \frac{1}{2}gt^2 =) 0.18 = 0.5 \times 9.81 \times t^2$ giving $t = 0.19$ (s) $(x = vt =) 0.42 \times 0.19 = 0.08$ (m) distance = $\sqrt{(r^2 + x^2)} = \sqrt{(0.0020 + 0.0064)}$ = 0.092 (m)	C1 C1 C1 A1	alt: projectile motion: $x = vt$, $y = \frac{1}{2}gt^2$ $y = \frac{1}{2}g(x / v)^2$ ecf (b)i for v; $x^2 = 2yv^2/g$ = $2 \times 0.18 \times 0.422/9.81$ Examiner's Comments About half of the candidates found the time for the ball to fall to the bench. Most then managed to find the horizontal distance from the point of release, but half forgot that the point of reference in the question was the centre of rotation so failing to complete the calculation.

Question			Answer/Indicative content	Marks	Guidance
	d		T increases or string stretches or angle θ increases to provide / create a larger centripetal force	M1 A1	allow mv^2/r or $mr\omega^2$ in place of <i>centripetal force</i> causality must be implied to gain the A mark Examiner's Comments About half of the candidates appreciated that the tension in the string increased or that the angle of the string to the vertical increased. Most answers gave the impression that the <i>centripetal force</i> was a <i>real</i> force rather than its provision being necessary for the ball to follow a circular path
			Total	12	
15			C	1	
			Total	1	
16			D	1	Examiner's Comments The tension for both wires is the same, yet wire X has half of the cross-sectional area. This means the stress for X will be twice that of Y. Strain = stress/Young modulus, so with half of the stress and (120/200) of the Young modulus, the strain for X will be $2 \times (200/120)$ or times that of the strain for Y. The original lengths for X and Y are the same, so the extension of X will be 3.3 times that of Y.
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
17	a		Maximum energy is transferred between tower (driver) and sphere when sphere (driven) is at/close to the natural frequency of the tower or in this forced <u>oscillation/resonance</u> situation	B1	allow causes maximum damping of the tower or maximum amplitude of the sphere/AW
				B1	allow AW e.g. sphere must be driven close to/at the natural/resonance frequency of the tower Examiner's Comments The answers gave a clear indication as to how well the candidates understood a resonance situation. Many omitted to explain which of the three oscillating elements were acting as drivers and which were driven. The candidate who wrote the answer (exemplar 3) shown here has some understanding of the situation but has failed to communicate it clearly to the reader. Exemplar 3
					because the maximum amplitude is produced when the system is resonant which is when the natural frequency is equal to the driving frequency and the natural frequency is 0.15 Hz so resonant when driving freq = 0.15 Hz [2] The ball was often quoted as just acting against the tower to reduce the amplitude rather than using the clue at the end of the initial paragraph about the energy drawn from the tower being absorbed by the dampers. Hence the requirement for the ball to be given a large amplitude or absorb the maximum amount of energy.
	b	i	$\omega^2 = k/m$ or $(2\pi f)^2 = k/m$ or $kA = m\omega^2$ $k = (m\omega^2) = 6.6 \times 10^5 \times (2\pi \times 0.15)^2$ or $(k = m\omega^2) = 6.6 \times 10^5 \times 0.05/0.056$ $k = 5.9 \times 10^5 \text{ (N m}^{-1}\text{)}$	C1 M1 A1	allow ω or $\omega^2 = 0.88$ or 0.89 quoted from (a) ecf value of A from (a) as this is a 'show that' question some definite evidence of working must be shown. not $k = 6 \times 10^5$ allow answer to 2 or more SF.

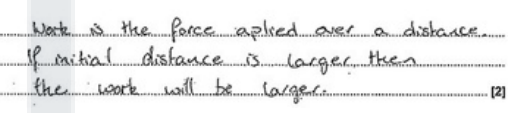
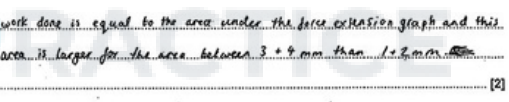
Question			Answer/Indicative content	Marks	Guidance
		ii	$E = \frac{1}{2}kA^2 = 0.5 \times 5.9 \times 10^5 \times 0.712$ $E = 1.5 \times 10^5 \text{ (J)}$	C1 A1	allow value from (c)(i) or 6; or $a = (k/m)A$, $F = ma$, $E = \frac{1}{2}FA$ accept 1.48 to 1.51 or value from ecf special case: give 1/2 for $E = 3(.0) \times 10^5$ (J) where it is clear that 2k has been used as the spring constant Examiner's Comments The exercise in this section completed successfully by most candidates was to perform standard calculations stating correct formulae and showing clear working to determine the required quantities. The example (exemplar 4) shown here is of a typical neat script. The most common error was to forget to square quantities in part (ii) or to use the amplitude calculated part (a) rather than the figure given in the stem of this part. Exemplar 4  Handwritten work for Exemplar 4: $0 = -\frac{k}{m}x \quad \frac{0.05}{0.056} \times (6.6 \times 10^3) = k$ $0.05 = -\frac{k}{(6.6 \times 10^3)} (0.056) \quad k = 5.89 \times 10^5 \text{ N m}^{-1}$ $\omega = 2\pi f \quad \frac{0.050}{(0.30\pi)^2} = x$ $\omega = 0.30\pi \quad x = 0.056$ $a = -\omega^2 x$ maximum displacement = 0.056 m [3] (14) Evaluate the work done by the force as a function of the displacement x. Second variant for M1 given with evidence of working. $E = \frac{1}{2}kx^2$ $E = \frac{1}{2}(5.89 \times 10^5)(0.71)^2$ $E = 148529.46$ $E = 1.49 \times 10^5 \text{ energy transferred} = 1.49 \times 10^5 \text{ J}$
		Total		7	

Question			Answer/Indicative content	Marks	Guidance
18			D	1	<u>Examiner's Comments</u> The Young modulus is found by calculating the initial gradient of the material's stress- strain graph. The initial portion appears to be a straight line from the origin to the point (0.1, 120). The units on this graph are megapascals and %. This means the co- ordinates of the chosen point are in fact $(0.1 \times 10^{-2}, 120 \times 10^6)$. Many candidates forgot to convert the strain into a decimal and left it as a percentage. Their answer was a factor of a 100 out, ie answer B. The correct answer is $120 \times 10^6 / 0.1 \times 10^{-2} = 1.2 \times 10^{11}$ Pa. This is answer D.
			Total	1	



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Question			Answer/Indicative content	Marks	Guidance
19	a		The maximum (tensile) stress a material can withstand (before it breaks)	B1	<u>Examiner's Comments</u> Ultimate tensile strength is the maximum stress a material can withstand without breaking or failing. The most common incorrect answer included descriptions of force rather than stress.
	b	i	Use a micrometer / (Vernier) caliper Measure the diameter along its length in different directions (to ensure uniform cross-section) AW	B1 B1	Allow either along length or in different directions
		ii	1. Any value between 0 and 2.0 (kN) 2. Any value between 2.0 and 2.2 (kN)	B1 B1	
		iii	$k = \text{gradient}$ $k = 5.0 \times 10^5 \text{ (N m}^{-1}\text{)}$	C1 A1	Allow 1 mark for 5.0×10^n ; $n \neq 5$ Allow $5 \times 10^5 \text{ (N m}^{-1}\text{)}$
	c		$(A =) \pi \times 0.00212 \text{ or } 1.39 \times 10^{-5} \text{ (m}^2\text{)}$ $(\sigma) = 2200 / \pi \times 0.00212$ $\sigma = 1.6 \pi 10^8 \text{ (Pa)}$	C1 A1	Allow 1 marks for $4(.0) \times 10^7$; diameter used as radius Answer is $1.59 \times 10^8 \text{ Pa}$ to 3sf

Question			Answer/Indicative content	Marks	Guidance
	d		Greater area under the graph (from 3 mm to 4 mm) / greater average force from (3 mm to 4 mm)	B1	ORA Allow: labelled/annotated diagram
			Mention of work done = average force x distance or work done = area under graph	B1	Allow energy (transferred) instead of work done Allow 2 marks for arguments including reference to $W = \frac{1}{2} kx^2$ and constant k and greater average x <u>Examiner's Comments</u> This question is best answered by referring to the graph in the question. Exemplars 1 and 2 indicate the difference between a low level and a high level response. Exemplar 1  This exemplar shows a response that contains broadly true statements yet is only loosely linked to the context and so scored zero marks. Exemplar 2  This response is clearly and specifically about this context and uses the graph as supporting evidence. The link to the graph is that the area gives the work done and that a larger area for one region means a larger amount of work done for that region.
			Total	11	

Question			Answer/Indicative content	Marks	Guidance
20	a	i	points on the line read to the nearest half square size of triangle is greater than half the length of the drawn line and $\Delta y / \Delta x$	B1	Allow Δy for $y_2 - y_1$ and Δx for $x_2 - x_1$
				B1	$\Delta x \geq 0.1625$
		ii	$\left(\frac{9.81}{0.12}\right) = 81.75$	C1	Allow ECF from (a)(i)
			82 N m ⁻¹ given to 2 or 3 significant figures	A1	Allow 81.8 N m ⁻¹ Note POT must be correct for given unit Allow kg s ⁻²
	b	i	steepest or shallowest line that passes through all the error bars	B1	
		ii	gradient determined: 0.10 m kg ⁻¹ or 0.13 m kg ⁻¹	B1	Allow ECF from (b)(i)
		iii	$\Delta \text{gradient} (0.13 - 0.12 \text{ or } 0.12 - 0.10)$	C1	Allow ECF from (b)(i) and (ii)
			$\frac{\Delta \text{gradient}}{\text{gradient}} \times 100 = 8.3\% \text{ or } 17\%$	A1	Not 10% without justification Examiner's Comments
			OR $\Delta k (82 - 75 \text{ or } 98 - 82)$	C1	In this question, most candidates clearly identified the points on the line that were to be used for the gradient calculation. High achieving candidates clearly showed their working when determining the percentage uncertainty.
			$\frac{\Delta k}{k} \times 100 = 8.5\% \text{ or } 20\%$	A1	
			Total	8	
21			C	1	
			Total	1	
22			D	1	
			Total	1	
23			D	1	
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
24	a		Arrow vertical down <u>and</u> an arrow opposite to the frictional force. Both arrows labelled correctly.	M1	Allow weight / mg / W for the downward arrow and tension / T / 'force in rod' / 'force in tow bar' / 'driving force' for the 'upward' arrow
				A1	
	b		($W_s =$) $1100 \times 9.81 \times \sin 10^\circ$ or $1100 \times 9.81 \times \cos 80^\circ$ ($W_s = 1874 \text{ N}$ or 1900 N)	C1 A0	Allow g instead of value
	c		force = $1900 + 300$ force = 2200 (N)	A1	Allow $1870 + 300 = 2170 \text{ (N)}$
	d		(distance =) $120 / \sin 10^\circ$ or 691 (m) (work done =) 2200×691 work done = $1.5 \times 10^6 \text{ (J)}$	C1 C1 A1	Allow ECF from (c) Allow ECF from an incorrect attempt at first mark.
	e		($A =$) $\pi \times 0.0062$ or $1.1 \times 10^{-4} \text{ (m}^2\text{)}$ (stress =) $\frac{2200}{\pi \times 0.0062^2}$ and $2.0 \times 10^{11} = \frac{\text{stress}}{\text{strain}}$ $x = 4.8 \times 10^{-5} \text{ (m)}$	C1 C1 A1	Allow ECF from (c) Allow $x (=FL/EA) = \frac{2174 \times 0.5}{2.0 \times 10^{11} \times 1.1 \times 10^{-4}}$ Allow 2 marks for 1.2×10^{-5} ; $1.2 \times 10^{-2} \text{ m}$ used as radius Allow answer between 4.7 and $5.1 \times 10^{-5} \text{ (m)}$

Question			Answer/Indicative content	Marks	Guidance
			Total	10	



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